

A and B are models describing similar and related RNA families within a larger space of all RNA sequences, R.

Sequences a_i are members of A; sequences b_i are members of B

`cmemit(A)` → set of sequences, members of A

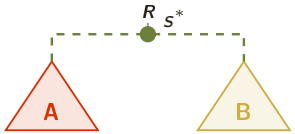
`cmsearch(`a_i`, B)` → log-odds bit score that a_i is part of B

`dinucshuffle(`a_i`)` → random sequence set with the same dinucleotide counts as a_i

How related are A and B?

1. Optimal sequence

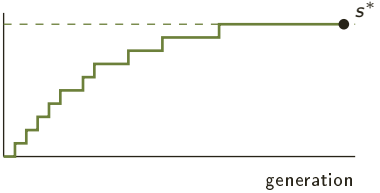
node joining A and B in R



s^*: node connecting A and B in R



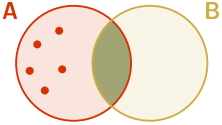
best score so far



optimal sequence s^* and its `cmsearch` scores

2. Median score

overlap of A and B, empirical p-value

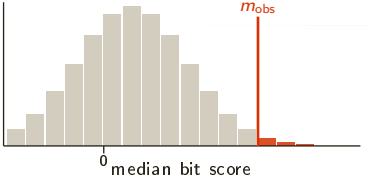


overlap ≈ median `cmsearch`(a_i, B)

Observed: $a_i \leftarrow$ `cmemit`(A),
m_{obs} = median `cmsearch`(a_i, B)

Null: null$_j$ = `dinucshuffle`(a_i), $j = 1 \ldots n$,
m_j = median `cmsearch`(null$_j$, B)

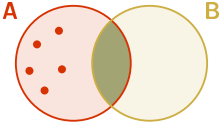
null medians m_j



median score m_{obs} and empirical p-value
$p = \#\{m_j \geq m_{\text{obs}}\}/(n+1)$

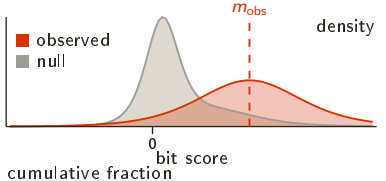
3. Score distribution

same overlap, compared by KS test

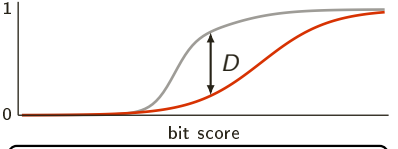


m_{obs} = median `cmsearch`(a_i, B)

`cmsearch`(a_i, B) vs null, whole distribution



cumulative fraction



median score m_{obs} and KS statistic D and p-value
$D = \max |F_{\text{obs}} - F_{\text{null}}|$